

**Solution**  
**MAGNETISM-03**  
**Class 10 - Science**  
**Section A**

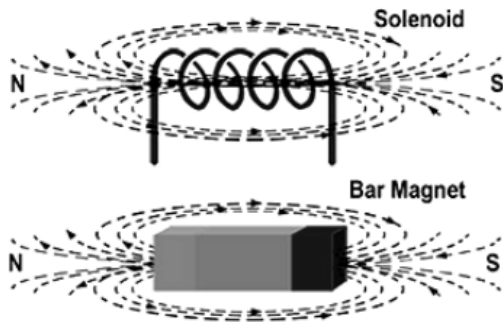
1. **(a) D**  
**Explanation:** Magnetic field emerges from north pole to south pole and has only one direction, thus no two field lines overlap.  
A wire with red insulation is usually the live wire of electric supply.  
So, all statements are incorrect.
2. **(a) Statement (c) is incorrect.**  
**Explanation:** The pattern of the magnetic field associated with a current carrying solenoid is different from the pattern of the magnetic field around a bar magnet.
3. **(b) If magnetic field lines are parallel and equidistant, they represent zero-field strength**  
**Explanation:** This statement is false about the magnetic field lines that if magnetic field lines are parallel and equidistant then they represent zero-field strength because if they are parallel and are at an equal distance then they have a uniform magnetic field and don't have zero magnetic field strength.
4. **(c) A and B**  
**Explanation:**  
A. A fuse is made of an alloy of tin and lead because it is important that the device acts without heating too much.  
B. A switch in the electrical circuit is always connected to the live wire so that the socket/appliance is not live when it is switched off.  
C. Alternating current is defined as the flow of charge that changes direction periodically.  
D. Higher is the value of current flow, stronger is the magnetic field  
So, Statement A and B are correct statement.
5. **(d) All of these**  
**Explanation:** Overloading takes place when the live wire comes into direct contact with the neutral wire (This occurs when the insulation of wires is damaged or there is a fault in the appliance). In such a situation, the current in the circuit abruptly increases. This is called Short-circuiting. Overloading is caused by connecting too many appliances to a single socket.  
So, all statements are correct.
6. **(a) B and C**  
**Explanation:** A. A fuse wire is a safety wire connected in series with the live wire, in case of any large current supply or malfunctioning in the electric connections it melts and breaks the electric circuit.  
B. A fuse wire is connected in series so as to pass the current totally through the fuse. It is not connected in the end of the circuit.  
C. The relation between thickness of a wire and its current carrying capability is that, more the thickness more the current a wire can carry. So fuse wires are made of sufficient thickness. Means thinner the fuse wire, lesser is its current capacity.  
D. The fuse needs to be rated slightly higher than the maximum operating current of the protected device, so that the fuse won't blow when the equipment is used.  
Statement B and C are not correct about electric fuse.
7. **(d) In India, the AC changes direction after every  $\frac{1}{50}$  second**  
**Explanation:** The frequency of Alternating current in India = 50 cycles/second  
Therefore, the time period of the alternating current  $\frac{1}{50}$  second  
Since AC changes the direction of current after every half time period. Therefore AC will change current in  $\frac{1}{100}$  seconds.
8. **(c) No two magnetic lines of force can intersect each other.**

**Explanation:** No two magnetic lines of force can intersect each other.

9. (a) The pattern of the magnetic field associated with the solenoid is different from the pattern of the magnetic field around a bar magnet

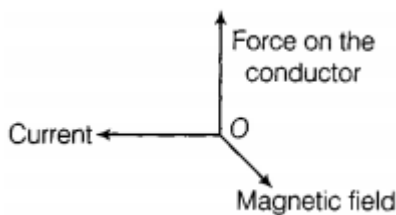
**Explanation:**

The pattern of the magnetic field associated with the solenoid is the same as the pattern of the magnetic field around a bar magnet.

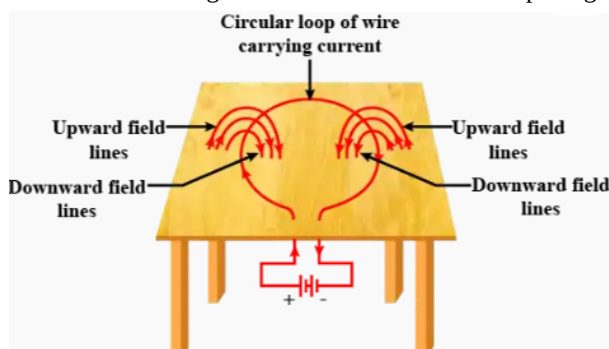


### Section B

10. The strength of the magnetic field is directly proportional to the relative closeness of field lines, therefore, region A has a stronger magnetic field.
11. A solenoid is a long coil containing a large number of close turns of insulated copper wire.
12. Place a compass near the wire, if the needle gets deflected then it indicates the presence of the magnetic field around the wire.
13. Near the poles of the magnet.
14. The strength and the direction of magnetic field at the given point.
15. The divergence of magnetic field lines indicates the decrease in strength of the magnetic field near the ends of the solenoid.
16. The direction of magnetic field is perpendicular to both current and force on the conductor. (according to Fleming's left-hand rule).

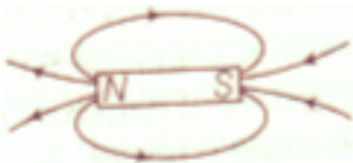


17. The type of core used in the electromagnet is soft iron core.
18. Every current-carrying conductor has a magnetic field around it.
19. The phenomenon is Electromagnetic Induction.
20. In the following diagram the current is flowing clockwise. If we are applying right hand thumb rule to the left side of the loop then the direction of magnetic field lines inside the loop are going into the table while outside the loop they are coming out of the table.

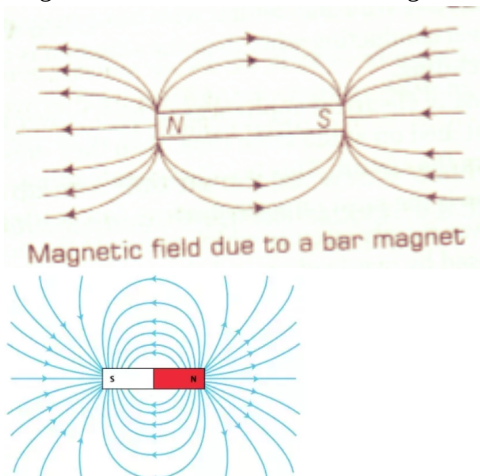


21.  $f = 50 \text{ Hz}$  and  $T = f^{-1}$   
so,  $T = 1/50 \text{ sec}$   
or,  $T = 0.02 \text{ sec}$   
Now, in 0.02 sec, the current changes its direction = 2 times  
therefore, in 1 sec, the current changes its direction =  $2/0.02 = 100$  times
22. The shape of magnetic field lines around the straight current carrying conductor is in the form of concentric circles.
23. The direction of magnetic field (and magnetic field lines) gets reversed on changing the direction of flow of current in a straight conductor.

24. The circuit B is dangerous even if fuse blows because the appliance still remains connected to the high potential of live (L) wire.  
If a person touches the faulty appliance, he will get an electric shock.
25. All the lamps should be connected in parallel.
26. Two safety measures used are:
- A safety fuse of proper rating. A fuse in a circuit prevents damage to the appliances and circuit due to overloading or short circuiting.
  - Earth wire prevents possible shock when live wire incidentally touches the body of appliance.
27. The two magnetic field lines can never intersect each other because two tangents can be drawn from that point of intersection which will give two directions of the field from the same point, which is impossible.
28. The nature of the coil is circular in form of loop.
29. A solenoid
30. a. 1 → Earth (E), 2 → Neutral (N) and 3 → Live (L).  
b. Terminal 1 is connected to the metal body of the appliance.
31. Fuse wire is fitted in a porcelain casing because porcelain is an insulator of electricity.
32. The property by virtue of which a magnet attracts certain metals such as iron, cobalt, nickel etc., is termed as magnetism.
33. The direction of the force will be reversed.
34. A piece of magnetic material can be magnetized by rubbing the north pole of a magnet on the magnetic material.
35. The earth wire carries the charges develop on the appliances immediately to earth, thus saving the life of the human.
36. If a swimmer swims in the direction of current, facing the magnetic needle, then the north pole of the magnetic needle deflects towards his left hand i.e., west and the south pole towards his right hand i.e., east.
37. According to the right-hand thumb rule, the curve of the fingers represents the direction of the magnetic field. If the wire is grasped in the right hand then the direction of magnetic field will be from North to South.
38. Strength of magnetic field is directly proportional to the strength of current flowing in the wire.
39. The poles of a magnet are marked in the figure as we know that out side magnet field lines moves  $N \rightarrow S$  and inside the Magnet field lines moves  $S \rightarrow N$ .

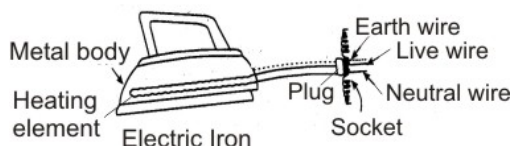


40. **FUSE** must be used to prevent shocking, overloading and short-circuiting in domestic appliances.
41. The magnetic field lines are closed curves because magnetic field lines originate from the north pole of a magnet and end at its south pole and inside the magnet, it is directed from south pole to north pole.
42. Magnetic field lines around the bar magnet in a region can be represented as given below -



43. The degree of closeness of magnetic field lines near the poles represents that the magnetic field is stronger near the poles.
44. The space or region around a magnet in which the force of attraction or repulsion due to the magnet can be detected is called the magnetic field.
45. i. An electric fuse is a safety device which is used to limit the current in an electric circuit. The use of a fuse safeguards the circuit and the appliances connected from being damaged.  
ii. An electric switch is an on-off device for current in a circuit or in an appliance.
46. No, a magnetic field exerts a force only on a moving charge.

47. An electromagnet is a soft metal core made into a magnet by the passage of electric current through a coil surrounding it.
48. A compass is a device used to show magnetic field direction at a point. It consists of a tiny pivoted magnet usually in the form of a pointer which can turn freely in the horizontal plane.
49. In order to work an electric apparatus, we require two wires, one live wire and the other neutral. Due to wear and tear or due to excessive heating, sometimes live wire touches the body of the apparatus. Thus we get shock when we incidentally touch the apparatus. To avoid the risk of electric shock, the metal body of the electric appliance is earthed i.e. a wire is connected to the cover of vessel on one side and to the earth point in three pin plug. The one end of the earth point is buried deep in the earth. Earth wire is usually given green or green-yellow colour, the live wire is usually red or brown while black or light blue wire acts as a neutral wire.



50. **The strength of magnetic field produced by a current carrying solenoid depends on:**

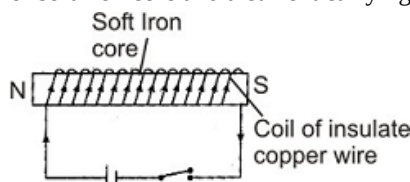
- The number of turns in the solenoid:** Larger the number of turns in the solenoid, greater will be the magnetism produced.
- The strength of current in the solenoid:** Larger the current passed through solenoid, stronger will be the magnetism produced.
- The nature of core material used in making solenoid:** The use of soft iron rod as core in a solenoid produces the stronger magnetism.

51. **Safety Fuse or fuse :** Usually the wire chosen for electric circuit are such that these allow a certain maximum current to pass through them without excessive heating of the circuits. However, incidentally there is a short-circuiting or over-loading, the current exceeds this maximum permissible value. The wires may get over-heated and catch a fire. Sparking at the points of short-circuit may also cause fire. Many precautions and safety measures are taken to protect the circuits against damage due to over-heating. All wires used in electric circuits are coated with layer of insulating materials. In addition these are coated with rubber or plastic layer.

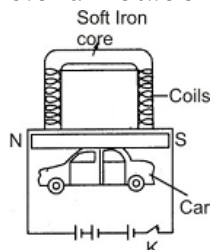
The most important safety device used these days is safety fuse or fuse. Fuse is a piece of wire of a material with a low melting point. Good fuse wire is always made of pure tin but cheaper variety is made of alloy of tin and copper or tin and lead (63% tin and 37% lead)

Fuse is always connected to the live wire. When current of value more than maximum permissible is passed through the circuit, the fuse wire melts due to excessive heating. This way the circuit is broken to ensure safety of the circuit. It is due to this fact that the fuse is usually called safety fuse. The thickness, length and material of the fuse wire depends upon the maximum current permitted through the circuit. For proper protection, a fuse of proper value is must. Fuse of improper rating is a curse instead of being a safety device.

52. Very strong electromagnets can be produced by binding an insulated copper wire on a soft iron core. Electromagnets work on the magnetic effects of electric current. We have studied in previous article that when a current is passed through a long solenoid, a magnetic field is produced. Now if a soft iron core is placed inside the solenoid, the strength of the magnetic field becomes very large. The reason for a large increase in magnetic field is due to the fact that iron gets magnetized by induction. The combination of soft iron core and a current carrying insulated copper wire wound over it is called an electromagnet.



A simple electromagnet is as shown in fig. To make an electromagnet, a soft iron core is taken and insulated copper wire is wound over it. The two ends of it are connected to a battery and a key. The electromagnet gets demagnetized when key is removed.

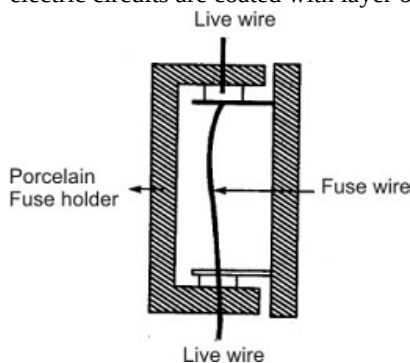


Strength of electromagnet depends upon

- i. Number of turns. Strength of electromagnets is directly proportional to the number of turns.
- ii. Current flowing. Larger the current flowing through the wire, stronger is the electromagnet. Strong electromagnets are used in cranes to lift the cars not obeying traffic rules.
- iii. Length of the air gap. Lesser the length of air gap between poles, stronger is the electromagnet.

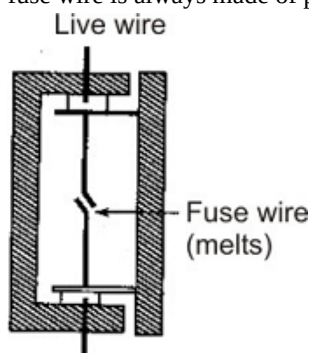
Air gap between the poles of a U-shaped electromagnet is small, hence it is very strong.

53. Usually the wires chosen for electric circuits are such that these allow a certain maximum current to pass through them without excessive heating of the circuits. If however, incidentally there is a short-circuiting or overloading, the current exceeds this maximum permissible value. The wires may get over-heated and catch a fire. Sparking at the points of short circuit may also cause fire. Many precautions and safety measures are taken to protect the circuit against damage due to overheating. All wires used in electric circuits are coated with layer of insulating materials. In addition, these are coated with rubber or plastic layer.



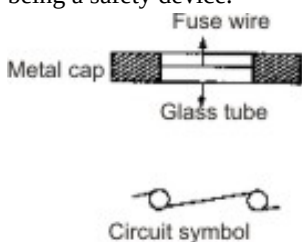
Intact fuse, current flows

The most important safety device used these days is safety fuse. Fuse is piece of wire of a material with a low melting point. Good fuse wire is always made of pure tin but cheaper variety is made of alloy of tin and copper or tin and lead (65% tin and 37% lead)



When excess current flows fuse wire melts and breaks

Fuse is always connected to live wire : When current of value more than maximum permissible is passed through the circuit, the fuse wire melts due to excessive heating. This way the circuit is broken to ensure safety of the circuit. It is due to this fact that the fuse is usually called safety fuse. The thickness, length and material of the fuse wire depends upon the maximum current permitted through the circuit. For proper protection, a fuse of proper value is a must. Fuse of improper rating is a curse instead of being a safety device.



Circuit symbol for fuse

54. **Precautions to be followed for using electricity are :**

- (1) All connects must be tight. Wires must be covered the proper insulation and of proper thickness. All joints must be covered with insulating tape. Defective switches should be immediately replaced.
- (2) Switch off all switches including main switch whenever there is a sparking or fire.
- (3) Fuse must be of proper rating and should always be connected to live wire.
- (5) Always put dry rubber shoes while repairing the circuit.
- (6) Fuses should be always connected to live wire. The earth wire must be connected to the body of electric appliance.
- (7) If in spite of all the precautions, a person gets shock due to accidental touching a live wire, one should try to provide such a person with support of some non-conducting material like wood, plastic or rubber. Never try to pull away person by your hand.

(8) Whenever repairs are needed, switch off main switch. If however, repairs need a direct handling of live wire, use of rubber gloves or rubber shoes or a plier with insulated handle is a must.

55. The activity to demonstrate that a current-carrying conductor experiences a force perpendicular to its length and the external magnetic field can be explained as follows:

**Activity:** To show the effect of magnetic field on a current-carrying conductor

Materials Required: For this, we need to take a small aluminum rod, a horseshoe magnet, battery, plug key, wires, and a stand.

- Suspend an aluminum rod horizontally from the stand and two wires at the ends of it are tied. The wires are connected to a Rheostat, battery and a key so that a circuit is completed,
- Place a horseshoe magnet in such a manner that the aluminum rod is between the poles of a magnet.

Assume that the above the aluminum rod is South pole of the magnet and below, the north pole of the magnet. Insert the plug key and current is supplied to the rod.

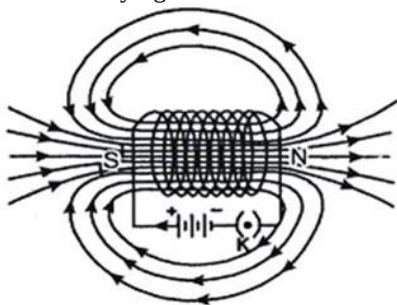
**Observation:** the aluminum rod is deflected towards the left direction

On changing the direction of the current, the rod is deflection in the right direction.

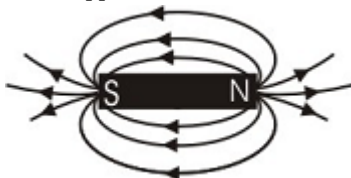
Hence, it demonstrates that a current-carrying conductor experiences a force perpendicular to its length and the external magnetic field

The direction of the magnetic field can find out with the help of Fleming's left-hand rule. Let current is moving in an anticlockwise direction, then the direction of the magnetic field will be in clockwise direction i.e. at the top of the loop whereas vice-versa in case of the clockwise direction of the current.

56. Solenoid is a coil of a number of turns of insulated copper wire closely wrapped in shape of a cylinder. Magnetic field around a current carrying solenoid is shown in fig.



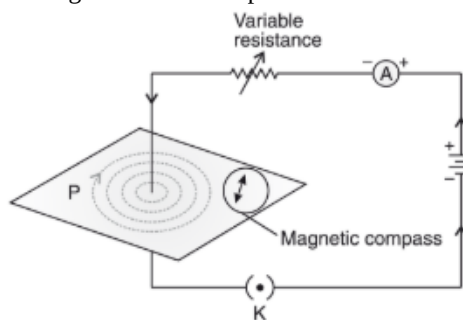
These appear to be similar to that of a bar magnet shown in below fig.



One end [right end] of solenoid behaves like north pole and the other end [left end] behaves like south pole. Magnetic field lines inside the solenoid are in the form of parallel straight lines. This means that the field is the same at all points inside the solenoid. When a soft iron rod is placed inside the solenoid, it behaves like an electromagnet.

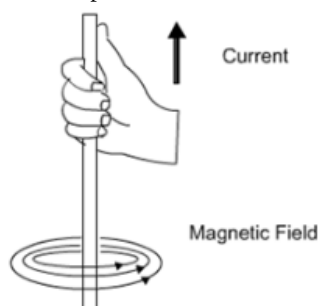
57. i. According to this rule, stretch the thumb, forefinger and middle finger of left hand such that they are mutually perpendicular. If the middle finger points in the direction of the magnetic field, the fore finger points in the direction of the flow of current, then the thumb points in the direction of motion.
- ii. Three characteristics of electric current use in our home are as follows:
- the current supplied in our homes is alternating current.
  - the current supplied in our homes is at 220 V.
  - the neutral wire and the live wire carry the current in our homes.
- iii. Fuse is a safety device used in a circuit to prevent damage due to overloading/short-circuiting. It protects the circuit by stopping the flow of any unduly high electric current. If current larger than the specified value flows through the circuit, due to Joule's heating effect the fuse wire melts and breaks the circuit.
- iv. When live wire touches the metallic appliance then electric current flows through the casing to the earth instead of the human body and thus we prevent ourselves from getting shocked. It is necessary to earth metallic casing of the appliance because it saves electrical appliance from burning and electric shock.

58. a. The magnetic field lines produced around a current-carrying straight conductor passing through cardboard is shown below.

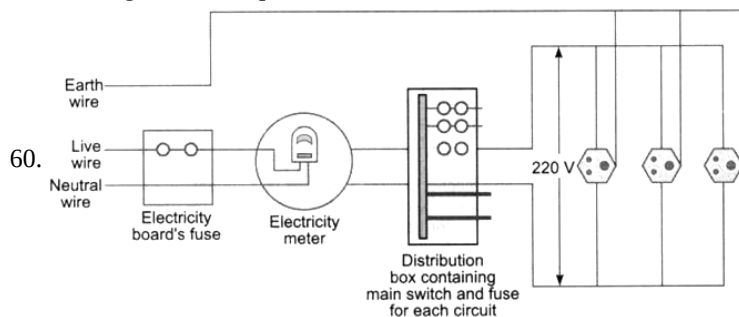


A right-hand thumb rule is applied to find the direction of these field lines. Imagine that you are holding a current-carrying straight conductor in your right hand such that the thumb points towards the direction of the current. Then your fingers will wrap around the conductor in the direction of the field lines of the magnetic field.

- b. When we move away from the straight wire, the deflection of the needle decreases which implies the strength of the magnetic field decreases. The reason is that the concentric circles representing the magnetic field around a current-carrying straight wire become larger and longer as the distance increases.
59. The pattern of the magnetic field lines of the magnetic field around a current-carrying straight long conducting wire are in a circular pattern in the form of concentric circles as shown in the below diagram:



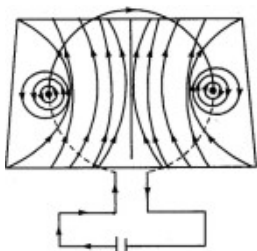
As depicted in the diagram, the direction of the magnetic field can find out by using the right-hand thumb rule which says that if we are holding a current-carrying conductor in the right hand such that the thumb will point towards the direction of the current. The fingers will wrap around the conductor in the direction of the field lines of the magnetic field.



60. A fuse in a circuit prevents damage to the appliances and the circuit due to overloading. Otherwise, the appliances or the circuit may be damaged.
- When current in the circuit exceeds the value of fuse rating, the fuse wire burns due to overloading. This causes a gap in the circuit and the current stops flowing in the circuit.
- This is done due to the reason so that the circuit or the appliances to be connected in the circuit continue functioning without any damage in future.
61. (1) Wires used for carrying current should be of proper current rating.  
 (2) Wire should be replaced by new wires of proper rating after every 3-4 years.  
 (3) Separate circuit should be there for heating appliance.  
 (4) Each circuit should have a fuse of proper rating.  
 (5) PVC of good quality should be used.  
 (6) Whereas wire of low current rating may be used for lighting electric bulbs, tubes, T.V. etc. wires of higher current rating should be used for A.C., heating appliances etc.  
 (7) Avoid using too many appliances in a single line, i.e. from a single output.  
 (8) Fuse or MCB of proper rating should be used.



62. **Magnetic field due to circular conductor:** When a current is passed through a circular coil in clockwise direction, the magnetic field will be as shown in figure.

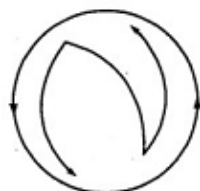


Here the current is passing clockwise when looked from front side. At the points where the conductor passes through horizontal plane, the lines of force are almost circular, their direction being given by right-hand grip rule. Near the central region, the lines are quite straight and at right angles to the plane of the coil.

Further, it is clear from the figure that all the lines enter at the nearer face and leave at the farther face. Evidently, the lines of force due to the current flowing in circular loop closely resemble those produced by a magnetized circular disc of steel of same boundary as that of the coil, so that one face of it is north and other a south pole. The polarity of any face of coil can be determined by remembering a simple rule known as clock rule. If the current round any face of the coil flows in an anticlockwise direction it behaves like a north pole and if the current is in clockwise direction, the face acts as a south pole.

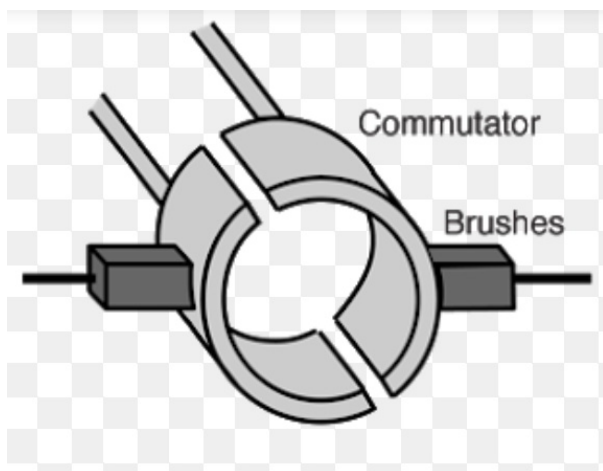


Clockwise current behaves like south pole.



Anti-clockwise current behaves like north pole.

63. a. **Fleming's left-hand rule:** Adjust your forefinger, middle finger and thumb of left hand in such a way that they are mutually perpendicular to one another. If the forefinger point in the direction of magnetic field, middle finger point in the direction of current then the thumb show the direction of force or motion on the current carrying conductor.
- Function of brushes:** Maintain contact between the coil and the external circuit.
  - Function of split rings:** Reverse the direction of current after each half rotation of the coil so that the coil can keep rotating continuously.

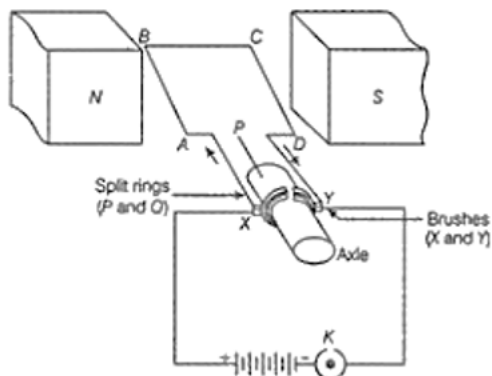


64. a. If the current is increased in the conductor the deflection of the compass needle increases. This is because the strength of magnetic field varies directly as the magnitude of the electric current, or the current passing through the wire. It indicates the presence of magnetic field.
- b. Right hand thumb rule states that, "If you imagine holding a current carrying wire in your right-hand with your thumb pointing towards the direction of electric current flow then the direction in which your fingers curl, gives the direction of lines of force of the magnetic field".

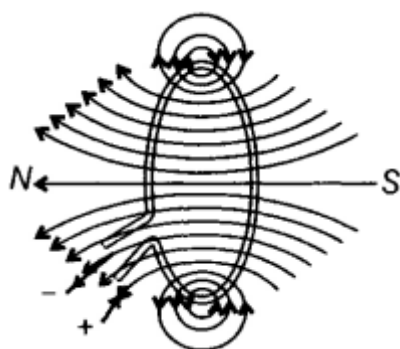


65. a.
  - Fleming's left-hand rule.
  - Adjust your hand in such a way that the forefinger points in the direction of magnetic field and the centre finger points in the direction of current, then thumb gives the direction of force acting on the conductor

b. Electric motor.

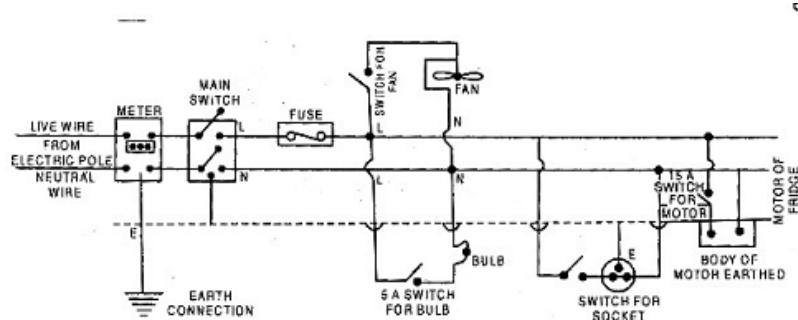


66. i. The magnetic field lines due to a circular coil are shown in the figure given below. At every point on a current carrying circular loop, the magnetic field is in the form of concentric circles around it. As we move away from it, the radii of the circle tend to increase. When we reach the center of the loop, the field appears to be a straight line.



- ii. a. No, it will not experience any force. As, magnetic field exerts force on a moving charged particle only.  
b. No, it will not experience any force because magnetic field exerts a force in perpendicular direction to motion of the particle.  
c. Yes, it will experience a force in a direction perpendicular to the direction of its own motion and the direction of magnetic field can be determined by Fleming's left hand rule.
67. There are two system of circuits :

- i. Tree system : Electricity supplied to us for domestic purposes is 220 V and A.C. The current alternates its polarity 50 times in 1s i.e., its frequency is 50 Hz., 15 A switches and 15 A sockets are used where heavy currents are to be drawn e.g., in hot air blower, motor, geyser etc. and 5 A switches, sockets are used for bulbs, tubes, radio, C.T.V., refrigerator fans etc. The main supply is delivered to houses using a three-core wiring called live, neutral and the earth. The live wire is red (or brown) in colour and brings current. It is dangerous to touch live wire with bare hands. Neutral is black (or light blue) coloured wire and is used as return wire. The third wire, green (or yellow) in colour is earth wire. Earth may be taken from pole or a metal plate sunk deep in the earth. It is safety measure and does not in any way affect the supply. Earth is usually connected to the body of the electric appliances. If at any time, the live wire incidentally touches the body of the electric appliance, the earth wire sends the current from the body of the appliances to the earth. Each appliance has a separate switch connected across live wire. The neutral reaches the appliance directly. All the appliances are connected in parallel so that switching off one circuit does not affect the other section. The parallel circuit also ensures that each appliance is connected to same voltage

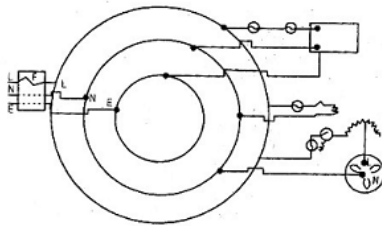


Advantages of tree system :

- Since all appliances are in parallel, each appliance gets optimum current.
- Since all circuits are in parallel, P.D. at the ends of each appliance remains the same.
- Each circuit is independent of the other. If there is a short circuiting in one circuit then only the fuse in that circuit gets melted. Power supply to the other circuit remains unaffected.
- Plugs and sockets to different amperage for different appliances are required.

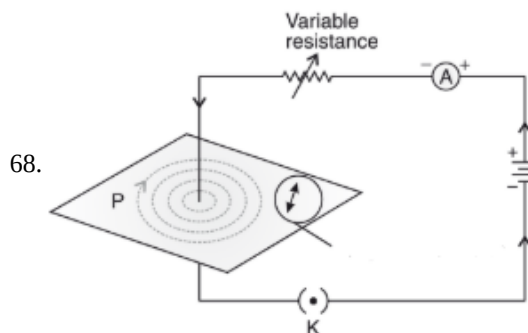
Disadvantages of tree system:

- Since all the circuits starts from main distribution board, therefore, longer lengths of the wire are required for each circuit. It increases the cost of wiring system.
  - Installation is fairly time consuming
- ii. Ring system of household wiring: Ring system consists of a ring-circuit starting distribution board and running around every corner and returns back to the distribution board again. There are two routes through which the current can flow through various appliances. Hence, a wire of lower rating may be used thus reducing the cost of installation.

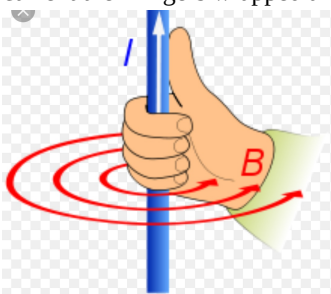


Advantages of ring system

- All appliances have separate fuse. In case of short-circuiting, only one fuse goes away.
- Its installation and maintenance is cheaper.
- Since ring feeds whole house, the length of the wire used is much smaller. It is cost-saving.



**Maxwell Right-hand Thumb Rule:** Hold a current carrying conductor in right hand such that thumb points the direction of current then fingers wrapped around the conductor. The fingers point in the direction of magnetic field lines of the magnetic field.



**Reason:-** The magnetic field strength decreases with increase of distance from the current carrying conductor.

There is an inverse relationship between field strength and distance from current carrying conductor.

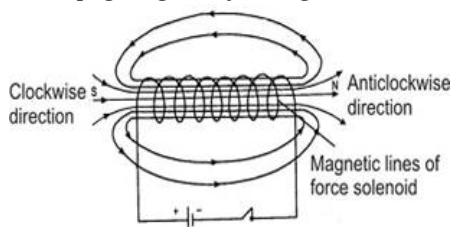
$$\text{Field strength} \propto \frac{1}{\text{distance of point from the current carrying conductor}}$$

|        |                                                                                                                                             |                                                                                                                                                                                           |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 69. a. | <b>I</b>                                                                                                                                    | <b>II</b>                                                                                                                                                                                 |
|        | Current (I) shows Direct Current (D.C.).                                                                                                    | Current (II) shows Alternating Current (A.C.).                                                                                                                                            |
|        | Current D.C flows in one direction only. It is called a direct current. The magnitude and direction of the flow of current remain the same. | Current A.C reverses direction after equal intervals of time. It is called alternating current. The magnitude and direction of current change continuously at definite intervals of time. |
|        | The magnitude of current in D.C does not become zero with the passage of time.                                                              | The magnitude of A.C becomes zero after a regular time interval.                                                                                                                          |

b. Source of (I) D.C. → A cell, battery, D.C. generator,

Source of (II) A.C. → A.C. generator.

70. A solenoid is a long circular coil containing a large number of close turns of insulated copper wire. When a electric current is passed through the solenoid, it produces magnetic field around it as shown in fig. Magnetic field produced by a current carrying solenoid is similar to the magnetic field produced by a bar magnet. As is clear from the figure, the lines enter from the left side and leave out from the right side. If we look from left side, the current appears to be passing in the coil in clockwise direction and hence it acts as a south pole according to clock rule. If the coil is viewed from right side, the current appears to be in anticlockwise direction. Hence, left-hand side face behaves as if this were a north pole. If the coil is left free, it will point South and North. Since the current in the turns of the solenoid flows in the same direction, the magnetic field produced by each turn of the solenoid adds up, giving a very strong resultant field inside the solenoid. Hence, a solenoid may be used in making electromagnets.



Strength of the magnetic field produced depends upon the following three factors :

Number of turns: Large the number of turns, stronger will be the magnetic field produced.

Strength of the current in the solenoid: Larger the current, stronger will be the magnetic field produced.

Nature of core of solenoid: The strength of the field depends upon the core on which the coil is wound. For air core, field is very mild whereas for iron-core, the field is very strong.

71. Take a straight conducting wire AB which passes through a horizontal cardboard. The ends of the wire are connected to a battery as shown in fig. When the key is closed, the current flows through the wire from B to A as shown in fig. (a), it produces magnetic field around it

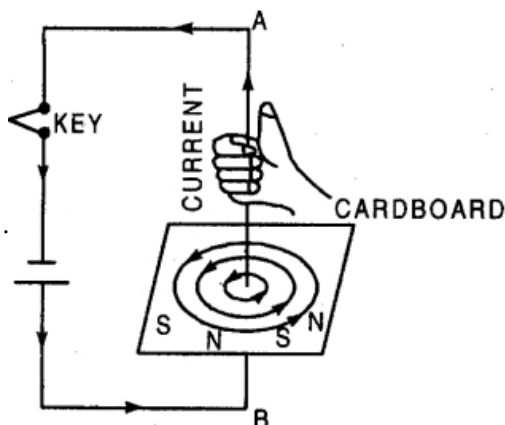
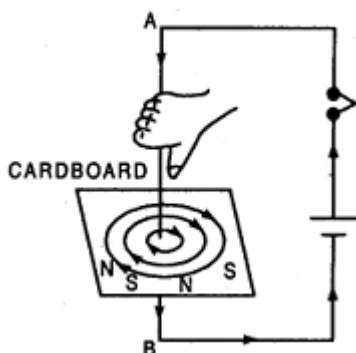
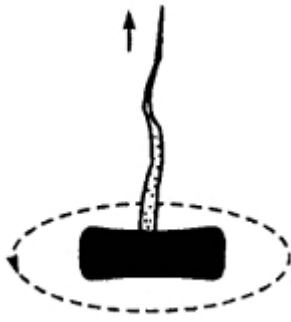


fig. (a)

The magnetic lines of force can be drawn with the help of a compass needle. The magnetic lines of force can also be visualized by sprinkling iron filings on the cardboard. On tapping the cardboard sheet, the iron filings arrange themselves in circles around the wire. The direction of the field is indicated by compass needle (a) The direction of magnetic field is given by right hand grip rule and by right hand cork screw rule.

Right hand grip rule is stated below : Grasp the wire in the right hand so that the thumb points along the wire in the direction of current, the fingers will then point in the direction of magnetic field.





Right hand cork-screw rule : Imagine a right handed cork-screw to be lying with its direction coinciding with the conductor carrying current and to be revolved so that it travels in the direction in which thumb rotates gives the direction of lines of force.

### Section C

72.
  - i. (d) all of the above
  - ii. (c) into the page
  - iii. (d) upward
  - iv. (b) electrical energy to mechanical energy
  - v. (d) vertically upward
73.
  - i. (d) same at all point
  - ii. (c) Oersted
  - iii. (c) Parallel to the axis of tube
  - iv. (d) The field consists of concentric circles centred on the wire.
  - v. (b) direction of current in electromagnet winding were reversed
74.
  - i. (d) in all plane around the magnet
  - ii. (c) originate from the north pole and end at its south pole
  - iii. (c) do not cross one another
  - iv. (d) There must be a relative motion between the coil of wire and a magnet
  - v. (c) If magnetic field lines are parallel and equidistant, they represent zero field strength